Guillaume Keifer U. Arenas

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EDUCATION

De La Salle University *2021 – expected 2027*
Bachelor of Science and Master of Science in Manufacturing Engineering and Management
CGPA: 3.58/4.00; Candidate for Cum Laude; DOST-Merit and ERDT Scholar

EXPERIENCE

Graduate Research Assistant, Taiwan Experience Education Program *Aug 2025 – Dec 2025; May 2026 – Jul 2026*
National Chin-Yi University of Technology, Taichung, Taiwan GitHub

- Built a mechanoluminescent elastomer grid prototype and a Python feature-extraction pipeline converting video trials into light-intensity features for modeling; co-authored "Preliminary Evaluation of Mechanoluminescent Lattice-Patterned Sensing Skin for Visuotactile Contact Differentiation," published in IEEE ICCT 2026.
- Delivered a MediaPipe-based knee-angle baseline (hip–knee–ankle landmarks), generating joint-angle time-series data adopted into an NSTC-funded project's motion-analysis workflow.
- Designed and fabricated a wrist-conforming sensor housing in Autodesk Fusion and built a low-cost automated 3-axis testing rig with a computer-vision marker-tracking pipeline to screen 4 candidate silicone elastomer blends.

Senior Officer *2024 – 2025*
Society of Manufacturing Engineers – DLSU Chapter

- Assisted in organizing events for the DLSU Department of Biomedical, Mechatronics, and Robotics Engineering, and documented 40+ students across plant visits in Osaka, Japan, curating 300+ photos into promotional material.

PROJECTS

GripVT: AI-Driven Wrist-Worn Sensor for Grip-Force Prediction (Master's Thesis) *May 2026 – Present*
Python, OpenCV, SciKit-Learn — submitted to IEEE Sensors Journal (under review) GitHub

- Engineered a wrist-conforming sensor housing (Autodesk Fusion) mounting a 9×11, 99-marker visuotactile sensing array against forearm curvature, enabling ergonomic, skin-conforming wear for continuous grip-force capture.
- Built the GripVT data-collection application (Python, CustomTkinter) covering calibration, live camera feed, and de-identified, crash-safe session storage; validated with a 379-test suite across 21 files (pytest).
- Re-engineered the computer-vision tracking pipeline for the new marker layout, resolving two scheduling bottlenecks to more than double throughput from 12.9 to 26.7 fps.
- Designed a leakage-safe, session-level nested cross-validation framework on a 69-session, cross-correlation-synchronized dataset, benchmarking Ridge, XGBoost, and MLP under equal Optuna tuning and MLFlow tracking; predicted grip force within ~6% of a wearer's own maximum strength on repeat wear and ~10% on unseen wearers.

Low-Cost Automated Visuotactile Sensor Testing Rig *May – Jun 2026*
Python, OpenCV, Ender 3 (G-code) — submitted to VTCA 2026 GitHub

- Engineered a real-time computer-vision pipeline (Python, OpenCV, SciPy) tracking 150+ markers/frame at ~30 FPS via Kalman filtering and Hungarian assignment, generating 1.6M+ structured measurements per run on a self-built rig (Ender 3 3D printer retrofitted with a load cell and camera, ~98% cost reduction vs. $10K–$14K commercial systems).
- Compared 4 elastomer blends via one-way ANOVA + Tukey HSD ($p<0.001$), identifying a statistically dominant formulation and delivering a data-driven material recommendation.

News Summarizer Bot *Apr 2026*
Python, LLM (BERT), Web Scraping, GitHub Actions GitHub

- Built an automated pipeline that scrapes news from 3 sources, summarizes articles with an LLM (BERT-based), and delivers digests via Telegram, scheduled through GitHub Actions cron jobs.
- Sustained 97.7% run uptime and a 90.8% article-summarization success rate across 88 automated runs and 1,500+ articles over 3 months of continuous operation.

Intelligent Quality Control System For Defect Detection and Classification in Packaging Cartons *Apr 2025*
Python, Arduino (C++), OpenCV, ESP32-CAM GitHub

- Assembled an ESP32-CAM, Arduino Uno, and conveyor-sorting hardware, then annotated and augmented a 170-image carton dataset to 510 images (408 train / 102 validation) to train a YOLO11n defect classifier achieving 94.8% recall, 87.2% precision, and 97.0% mAP50 across sealed, open, torn, and holed classes.
- Deployed a real-time Python/OpenCV detection pipeline over ESP32-CAM streaming to trigger Arduino-controlled servo sorting and LED defect indication, incorporating ANSI/PMMI B155.1-2023 and ISO13849-1:2023 safety-guarding considerations into the hardware design.

PCB Defect Detection Research *Oct – Dec 2023*
IEEE TENCON 2024, DOI: 10.1109/TENCON61640.2024.10902913

- Optimized a YOLOv8s PCB defect detector with TensorRT, achieving 98.5% mAP50 while reducing inference time ~0.9ms and model size ~39.3% versus baseline; published in IEEE TENCON 2024.

SKILLS

Programming & Data Analysis: Python, C++, SQL, MATLAB, Pandas, NumPy, EDA, statistical analysis
Machine Learning & CV: scikit-learn, TensorFlow/Keras, XGBoost/LightGBM/Ridge, Optuna, YOLO, OpenCV, MediaPipe, BERT
AI Tools: Claude Code, Prompt Engineering, LLM API Integration
Tools & Automation: Git/GitHub, GitHub Actions, Jupyter Notebook, CustomTkinter, pytest, Matplotlib/Seaborn